

WORKSHEET 29

Voting Systems

Suppose we have an election, and there are several candidates. How should we design a voting system to pick the winner?

The most common voting system is known as *first past the post*. In this voting system, every voter votes for exactly one candidate, and the candidate who collects the most votes wins.

When there are two candidates, first past the post is a great system. But we quickly find problems with it when there are three or more candidates, in the sense that it does not necessarily capture the will of the voters effectively.

To see this, we'll need to know not just the voters' top choices, but their complete preference profiles, i.e. their first choices, their second choices, their third choices, and so on. So, let's suppose that there are three candidates. We'll distinguish them by colors: we have orange, green, and purple candidates. Suppose that there are 9 voters, and here are their preference profiles:

Voter	1	2	3	4	5	6	7	8	9
First	●	●	●	●	●	●	●	●	●
Second	●	●	●	●	●	●	●	●	●
Third	●	●	●	●	●	●	●	●	●

115
128
40

PROBLEM 29.1. Who wins this election under the first past the post system?

PROBLEM 29.2. Now, suppose that orange weren't part of the election, so it's just between green and purple. Now who wins, with the same preference profiles?

PROBLEM 29.3. Which candidate is most deserving of winning the election, given these preference profiles?

Before we move on and look at other voting systems and the issues they can suffer from, let me offer a bit of advice. It may be tempting to vote for your favorite candidate, and perhaps doing so makes you feel good about voting for the candidate you support. But instead, you should treat voting not as a way of displaying support, but as a game, and when you play a game you should try to work out the best strategy, so as to lead to as good of an outcome as you can. Vote in such a way

as to maximize the expected outcome (or some other metric), and this might not be the same as voting for your favorite candidate.

Suppose we have three candidates, and again we'll call them orange, green, and purple. Given all the voters' preference profiles, we might imagine that one candidate ought to win if that candidate is preferred to each of the other two in a two-candidate election. We'll call such a candidate the *Condorcet winner*.

PROBLEM 29.4. Which candidate is the Condorcet winner in the election above?

PROBLEM 29.5. Does there exist a preference profile for which there is no Condorcet winner? Assume that the number of voters is odd, so that there cannot be ties.

We say that a voting system satisfies the *Condorcet criterion* if the system always picks the Condorcet winner when one exists.

Another possible voting method is called the *Borda count*. Suppose there are n candidates. Then each voter ranks the candidates from 1 to n , with 1 being the top choice and n being the last choice. Then we give each candidate n points for each voter who ranked em in first place, $n - 1$ points for each voter who ranked em in second place, and so on.¹

PROBLEM 29.6. Does the Borda count satisfy the Condorcet criterion?

No, Counter example of 100 candidates
101 voters
polarizing
candidate
2nd winner.

Another voting system that has gained popularity recently, and is now being used in some places in the United States, is *instant runoff voting*, also known as *ranked choice voting*. In instant runoff voting, voters rank the candidates from 1 to n . If any candidate gets the majority of the first place votes, then that candidate is declared the winner. If no candidate gets the majority of the first-place votes, then the candidate who received the smallest number of first-place votes is eliminated, and all votes for that candidate are transferred to the voters' next pick; in particular, the voters who ranked that candidate first now move on to their second-place pick. This elimination process repeats until one candidate gets the majority of the first-place votes.

PROBLEM 29.7. Suppose we have five candidates, called A, B, C, D, E , and 20 voters. Here are the number of voters with each preference profile:

¹There are other conventions for how to award points based on rankings as well.

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1st	A	B	C
2nd	B	A	B
3rd	C	C	A

Number	3	4	4	6	2	1
1 st	B	C	B	D	B	E
2 nd	C	A	D	C	E	A
3 rd	A	D	C	A	A	D
4 th	D	B	A	E	C	B
5 th	E	E	E	B	D	C

D wins runoff,
Q wins Condorcet
Need 40 x 10V ex.

Who wins this election under instant runoff voting?

PROBLEM 29.8. Does instant runoff voting satisfy the Condorcet criterion?

No, B wins w/o satisfy.

By now, we have seen a number of voting systems, as well as some flaws with all of them. So, you might wonder if there are any voting systems that are free from the sorts of defects we have seen. Unfortunately, this is not the case. Let's take a look at some properties we might like to have, or that we might like to avoid:

Unanimity: If every voter ranks candidate A first, then A wins.

Honesty: Voters cannot benefit by voting in a way that does not represent their preferences.

Pareto efficiency: If every voter prefers candidate A to candidate B , then candidate A finishes ahead of candidate B . (If only one winner is chosen, this means that candidate B cannot win in this case.)

Independence of irrelevant alternatives: Given two candidates A and B , the relative rankings of A and B cannot change when one or more new candidates enter the race.

Dictatorship: Voter v is a dictator if, for all preference profiles for all the voters, the winner of the election is v 's top choice.

Of these, the most confusing is independence of irrelevant alternatives. But refer back to Problem 29.2, where we see an example of the failure of independence of irrelevant alternatives, since the presence of orange changes the relative ranking between green and purple.

Here are two rather unfortunate theorems about voting systems:

THEOREM (Gibbard-Satterthwaite). Any voting system with at least three candidates that satisfies unanimity and honesty is a dictatorship.

THEOREM (Arrow). Any voting system with at least three candidates that is Pareto efficient and independent of irrelevant alternatives is a dictatorship.

The proofs of these theorems are a bit beyond the scope of this book. However, we'll at least look at a few ingredients that go into the proofs.

PROBLEM 29.9. Call a candidate *polarizing* if all voters rank em either first or last. Show that the voting system that is Pareto efficient and independent of irrelevant alternatives must rank a polarizing candidate either first or last.

PROBLEM 29.10. Suppose we have a set of votes in a voting system that is unanimous and honest, and that candidate A wins. Suppose that a single voter changes eir preference profile by moving candidate B up some number of places in eir ranking. Show that now only A or B can win the election.

PROBLEM 29.11. With the hypotheses as in the previous problem, show that if one candidate moves A up some number of places in the ranking, then A still wins.

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29.1: If each voter voted for their first preference, then the votes are: G-4 O-2 P-3, so Green wins.

29.2: Now, voters 5 and 6 would choose their secondary preference, purple. Tally: G-4 P-5, so Purple wins.

29.3: Give each first choice 2 votes, each second choice 1 vote, and each third 0 votes.

Tally: G-11, O-4, P-12, so Purple wins.
(4x2+1x2) (0x7+2x2) (6x1+3x2)

29.4: Here, Purple is a Condorcet winner. In 29.2, they beat Green, and against Orange they win every vote except for 5's and 6's.

29.5: Yes - here is an example:

Voter	1	2	3	4	5	6	7	8	9
First	G	G	G	P	P	P	O	O	O
Second	P	P	P	O	O	O	G	G	G
Third	O	O	O	G	G	G	P	P	P

Each party beats 1 other in a 1v1 - so no clear Condorcet win.

*A term for this is "polarization" (opposite beliefs.)

29.6: No. Take the example of 100 candidates and 101 voters.

51 voters pick candidate G first, and 50 voters pick G last.*

All 101 voters pick P second.

The Borda count for P is clearly higher (99.101 P vs 100.51 G), but

G wins the Condorcet with a majority in the first place, guaranteed to beat everybody else.

Borda = P $\neq$ Condorcet = G, so it does not satisfy.

29.7: Order of Runoff Voting:

- A is eliminated, no 1st place votes
- E is eliminated, 1 1st vote only
- C is eliminated, only 4 1st votes overtaken by B/D
- D wins overall by an 11-9 marginal majority

29.8:

Votes Candidates	5	3	4
1st	A	B	C
2nd	B	A	B
3rd	C	C	A

 Instant Runoff:

• B is eliminated

• A wins with 8-4 majority

Condorcet: B beats A 7-5 in a 1v1 and

B beats C 8-4 in a 1v1. B win $\neq$ A win, so this is a counterexample.

29.9: Say that every voter ranks this candidate 1st or every voter ranks this candidate last. By Pareto efficiency, this candidate must finish either first or last. Now, say another candidate enters the race. By independence of irrelevant alternatives, this candidate cannot change the first/last candidate's polarizing relationship with all other candidates, so the polarized candidate must stay first/last.

29.10: In this case, either A wins or B, recently upvoted, wins - not some other candidate C. By the honesty principle, only B can benefit from a change in votes. Therefore, because C cannot benefit from the upvote, C cannot win - only A or B.

29.11: Same setup as in the previous problem. By unanimity, A wins if everyone ranks A first - but nobody moved A up (A was already 1st.) By honesty, only A can benefit if someone moves A up in preference, and no candidate C can be benefitted by someone dishonestly demoting A. If A was already winning, and only A can be benefitted by an upvote due to honesty, then A is the only possible winning candidate.